IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

**COMMUNICATIONS: PHYSICAL LAYER**

Friday, 1 May 2026, 10:00 AM

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Part 1 Questions are Q1 and Q2.**
**Part 2 Questions are Q3 and Q4.**

*Answer 1 (one) question from Part 1, with each carrying a weighting of 50%.*
*Answer 1 (one) question from Part 2, with each carrying a weighting of 50%.*

*This is an OPEN book examination.*

**Any special instructions for invigilators and information for candidates are on page 1.**

| | | |
|---|---|---|
| Examiners responsible | First Marker(s) : | S. Lucyszyn, H. Merdan |
| | Second Marker(s) : | H. Merdan, S. Lucyszyn |

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**Special instructions for invigilators**

*This is an OPEN BOOK examination.*

**Special instructions for students**

*There are 4 (four) questions in total.*

*Part 1 Questions are Q1 and Q2*

*Part 2 Questions are Q3 and Q4*

*Answer 1 (one) question from Part 1, with each carrying a weighting of 50%.*

*Answer 1 (one) question from Part 2, with each carrying a weighting of 50%.*


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Q1.

a)

i) What are the characteristics of a Comms link? Briefly explain the importance of latency in real-time communication.

[2]

ii) Briefly explain how does optical communication enable very high-frequency transmission, and what are the advantages of using very high frequency in communication?

[2]

iii) Give two examples of commonly used optical frequency windows in optical communication and very briefly explain why some optical frequency ranges are preferable.

[3]

b)

An EM wave on below plane in Figure 1 is specified as $y = 18\cos[2\pi(5t - 1.8z)]\mu m$.

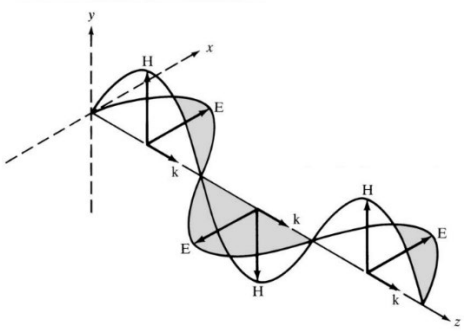



*Figure 1.*

i) Find the amplitude, the wavelength and the angular frequency of the given EM wave.

[3]

ii) Find the displacement of at time $t = 0$ and $z = 7\ \mu m$.

[3]

c)

i) A light travelling in air ($n_1 = 1.00$) is incident from the side on acetone ($n_2 = 1.356$) contained in a glass container ($n_3 = 1.52$) with parallel walls. The incoming ray from air to the surface of the glass container makes an angle of $\varphi_1 = 60°$ with respect to the normal, what is angle of refraction $\varphi_2$ in the acetone?

[2]

ii) Calculate the critical angle of the interface between a Polymethylmethacrylate (PMMA) surface ($n_1 = 1.489$) and water ($n_2 = 1.333$).

[2]

iii) For a light travelling from air ($n_{air} = 1.00$) into a silica fibre ($n_{core} = 1.333$; $n_{cladding} = 1.30$), calculate the numerical aperture.

[2]

d)

i) How does Rayleigh scattering loss vary with optical frequency?

[2]

ii) Briefly describe bending losses in optical fibres: where and how they tend to occur, and how they can be prevented.

[3]

iii) A communication link is needed between the Unit A and the Unit B of a factory which are 90 metres apart each other. The transmitter and receiver system can tolerate a maximum link loss of 12 dB with a safety margin 2 dB included. This communication line will need two connectors. As an engineer, you are allocated £600 for this project and asked to choose one of the options below. Based on considering the losses and cost, decide the most suitable option

| Option 1 | Option 2 |
|---|---|
| Glass optical fibre, with attenuation of 0.35 dB/km.<br>£2 per metre of glass optical fibre.<br>Connector loss: 0.3 dB/km per connector, £10 each.<br>Installation cost: £200 | Plastic optical fibre, with attenuation of 100 dB/km.<br>£0.80 per metre of plastic optical fibre.<br>Connector loss: 0.6 dB/km per connector, £4 each.<br>Installation cost: £80 |

[7]

iv) If the Unit A is moved to adjacent branch which is 1km away from where the Unit B is, explain if this would affect your choice regarding the signal losses. (Ignore the budget limitation in this case, and you do not need to recalculate the losses numerically, only estimation supported by a brief reasoning is enough.)

[3]

v) Silica Fibre A and Silica Fiber B are doped with 7.5% and 15% mole fractions of $GeO_2$. Estimate the ultraviolet absorptions at 0.9 μm and 1.2 μm for Silica Fiber A and B? From your results, briefly discuss how $GeO_2$ doping and wavelength affects ultraviolet loss.

[8]

vi) If the scattering loss is estimated as 0.11 dB/km for a silica-based fibre at 1.55 μm, calculate the refractive index of the silica fibre ($T_f = 1400\ K;\ \beta_T = 7 \times 10^{-11}\ {}^{m^2}/_{N};\ k_B = 1.38 \times 10^{-23}\ {}^{J}/_{K};\ p = 0.28$).

[8]

Q2.

a)

i) For the Dense Wavelength Division Multiplexing (DWDM) system using equal-frequency channel spacing, around 1550 nm, calculate the spectral bands for the channel spacing ($\Delta\nu$) of 50 GHz, 100 GHz and 200 GHz ($c = 3 \times 10^8$ m/s).

[6]

ii) In a DWDM system around 1550 nm with 200 GHz channel spacing, how many independent signal channels could be fit into the bandwidth between 1465 nm and 1625 nm?

[6]

iii) In a 16-channel DWDM system using uniform channel spacing is $\Delta\nu = 50$ GHz. In this system, the frequency $\nu_n$ correspond to the wavelength $\lambda_n$. E.g. $(\lambda_1, \nu_1\,;\ \lambda_2, \nu_2\ ;\lambda_3, \nu_3\,;\ldots\ldots;\lambda_{14}, \nu_{14}\,;\lambda_{15}, \nu_{15}\,;\ \lambda_{16}, \nu_{16})$. Assume the wavelength $\lambda_1 = 1535$ nm. Calculate the wavelength spacing between channel 4 and 5 ($\Delta\lambda_a$) and channel 14 and 15 ($\Delta\lambda_b$). From your findings ($\Delta\lambda_a$ and $\Delta\lambda_b$), what can be observed about using equal-frequency channel spacing instead of using equal-wavelength spacing in this wavelength band ($c = 3 \times 10^8$ m/s).

[10]

iv) Draw the schematic of the Wavelength Division Multiplexing (WDM) and briefly explain the role of the components on the WDM communication line schematic.

[6]

v) Describe briefly how WDM works and list advantages and limitations of WDM Systems.

[6]

b)




*Figure 2.*

i) Figure 2 above shows a simplified optical communication system. Decide which component/components can be placed at A, B and C from the below list.

- A Laser Diode
- A Light Emitting Diode
- An Erbium-Doped Fiber Amplifier
- An Avalanche Photodiode

[6]

ii) A factory needs to monitor temperature and vibration in below factory units to make sure that the environment is safe and suitable for manufacturing and factory operations. For each factory unit given below, briefly explain why optical sensors are better choice than electrical sensors.

a) Power unit that is near high-voltage equipment
b) Inside a chemical plant where close to explosive gases pipes.
c) Loading/transportation unit with cranes using strong electromagnets.
d) Extreme temperature area housing industrial furnaces.

e) Wastewater treatment unit which is an area with high humidity and significant exposure to industrial water.

[10]

Q3.

a) Power is transmitted at 2.45 GHz.

i) Calculate the Reactive Near Field distance for a theoretical point source radiator.

[2]

ii) Calculate the Radiative Near Field distance for a theoretical point source radiator.

[2]

iii) Calculate the Far Field distance for a theoretical point source radiator.

[2]

iv) Calculate the upper limit of the Rayleigh region for a quarter-wave monopole. What is another name for this region?

[3]

v) Calculate the upper limit of the Fresnel region for a quarter-wave monopole. What is another name for this region?

[3]

vi) Calculate the lower limit of the Fraunhofer region for a quarter-wave monopole. What is another name for this region?

[3]

b) Antennas can be polarized in different ways.

i) Very briefly, explain what is meant by a vertically polarized antenna and what kind of carrier modulation was it traditionally used for?

[4]

ii) Very briefly, explain what is meant by a vertically horizontally antenna and what kind of carrier modulation was it traditionally used for?

[4]

iii) What are the linear polarization mismatch losses, in dB, for 45° and 90° mismatch angles between the transmit and receive antennas.

[3]

iv) Why do Wi-Fi routers employ antenna diversity switches?

[2]

v) Very briefly, describe circular polarization and give one application in which it is used.

[4]

vi) Given a left-hand circularly polarized (LHCP) or anti/counter-clockwise polarized transmit antenna, what kind of polarization is required by the receiving antenna for minimal loss?

[2]

vii) Given a right-hand circularly polarized (RHCP) or clockwise polarized transmit antenna, what is the theoretical minimal loss is the receiving antenna is linearly polarized?

[2]

c) Consider a space-to-Earth wireless power transfer system, whereby a satellite's solar panel array is used to transmit microwave energy at 5.8 GHz down to Earth via a circular parabolic reflecting antenna (Dish). With reference to Figure 1, if the geostationary satellite is positioned 36,000 km directly above the Earth's 3 km diameter circular receiving rectenna array, calculate the following and include all the units:

i) The angle subtended by Earth's receiving rectenna array (shown as a red circle in Figure 1) from the satellite's transmitting antenna $\psi$.

[2]

ii) The optimum beam solid angle $\Omega_M$ for the satellite's antenna. *Hint: for large aperture antennas having a single pencil beam radiation pattern, this is given by $\Omega_M \approx \psi_\theta \psi_\phi$, where $\psi_\theta$ and $\psi_\phi$ are the -3 dB beam widths in the E- and H-planes, respectively.* Clearly state any simplifying assumptions.

[2]

iii) The directivity of the satellite's antenna $D_{TX}$.

[2]

iv) The ideal effective aperture area of the satellite's antenna $A_{TX}$.

[2]

v) The ideal diameter of the satellite's antenna $d_{TX}$.

[2]

vi) What is a typical efficiency for dish antennas and use your value to calculate the gain of antenna.

[2]

vii) Is the size of the dish practical. If not, suggest a possible solution.

[2]

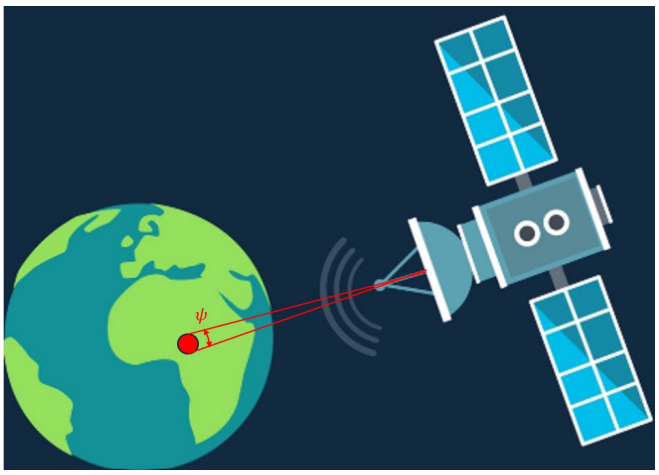



Figure 3.

Q4.

a) Consider millimetre-wave and sub-terahertz radiometric imaging.

i) State the mathematical expression for Planck's law in terms of zero-point energy, photon energy, thermally-induced energy and emissivity, and include all the units.

[5]

ii) From Plank's law, state the mathematical expression for the average noise power radiated and include all the units.

[2]

iii) What is emissivity and its practical implication?

[3]

iv) What approximation is needed to reduce Planck's law to the Rayleigh-Jeans law?

[4]

v) What are the considerations in choosing the most appropriate frequency bands of operation?

[8]

vi) What is the difference between passive and active radiometric imaging? Briefly explain the hardware considerations between the two. Comment on any health hazards, if any, between the two.

[8]

b) A radiometer has a pre-detection bandwidth of 20 GHz and a stand-off range of 2 m for detecting human body targets.

i) Calculate the average noise power radiated by target and include the units. State any assumptions for these calculations.

[4]

ii) Calculate the received power flux density from the target and include the units.

[2]

iii) The body scanner has a very large antenna, relative to its operational wavelength, deduce the antenna's equivalent noise temperature for the target. State any simplifying assumptions for these calculations.

[2]

iv) The body scanner has 5 dB noise figure, is cooled down to 77 K and has a post-detection integration time of 50 μs. Calculate the radiometric resolution.

[12]